

The Chain Rule & Directional Derivatives



The multivariable chain rule

- 1 Let $z = x^2 + y^2$, $x = \cos t$, $y = \sin t$. Find $\frac{dz}{dt}$.
 - 2 Let $w = xy + z^2$, $x = t$, $y = t^2$, $z = t^3$. Find $\frac{dw}{dt}$ at $t = 1$.
 - 3 Let $z = e^{xy}$, $x = t^2$, $y = 1/t$. Find $\frac{dz}{dt}$.
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Chain rule with two independent variables

- 4 Let $z = f(x, y)$ where $x = s + t$, $y = s - t$. Express $\frac{\partial z}{\partial s}$ in terms of $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
 - 5 For $z = x^2y$, $x = s + t$, $y = s - t$, find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$ at $s = 2$, $t = 1$.
 - 6 Let $z = f(x, y)$ where $x = r \cos \theta$, $y = r \sin \theta$ (polar coordinates). Write the chain rule expression for $\frac{\partial z}{\partial \theta}$.
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Implicit differentiation via the chain rule

- 7 Use the multivariable chain rule to find $\frac{dy}{dx}$ for the implicit curve $x^2y + y^3 = 8$, via the formula $\frac{dy}{dx} = -\frac{F_x}{F_y}$ where $F(x, y) = x^2y + y^3 - 8$.
 - 8 For the implicit curve $\sin(xy) = x + y$, find $\frac{dy}{dx}$ using $F(x, y) = \sin(xy) - x - y$.
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Gradient and directional derivative

- 9 Find the gradient of $f(x, y) = x^2y^3$ at $(1, 2)$.
 - 10 Find the directional derivative of $f(x, y) = x^2y$ at $(2, 1)$ in the direction of $\mathbf{v} = \langle 1, 1 \rangle$.
 - 11 Find the directional derivative of $f(x, y) = e^x \cos y$ at $(0, 0)$ in the direction of $\mathbf{v} = \langle 3, 4 \rangle$.
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The gradient points in the direction of steepest ascent

- 12 For $f(x, y) = 3x^2 - y^2$ at $(1, 2)$, find the direction of steepest ascent and the maximum rate of increase.
- 13 In what direction from $(1, 1)$ does $f(x, y) = xy$ decrease most rapidly, and at what rate?

- 14 Explain, using the formula $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = |\nabla f| \cos \theta$ (for unit $\mathbf{u}$), why the direction of steepest ascent is exactly ∇f itself.
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The gradient is orthogonal to level curves

- 15 Find ∇f at $(2, 1)$ for $f(x, y) = x^2 + 4y^2$, and state what geometric relationship it has with the level curve $f(x, y) = 8$ through that point.
- 16 Find the equation of the tangent line to the level curve $x^2 + 4y^2 = 8$ at $(2, 1)$, using the fact that $\nabla f = \langle 4, 8 \rangle$ is normal to it there.
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Application of the gradient

- 17 The temperature on a plate is $T(x, y) = 100 - x^2 - 2y^2$. A bug at $(2, 1)$ wants to cool off fastest. In which direction should it move?
- 18 Find the rate of change of $T(x, y) = 100 - x^2 - 2y^2$ at $(2, 1)$ in the direction of the unit vector $\langle \frac{3}{5}, \frac{4}{5} \rangle$.
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Related rates via the chain rule

- 19 The volume of a cylinder is $V = \pi r^2 h$. If r is increasing at 2 cm/s and h is decreasing at 3 cm/s, use the chain rule $\frac{dV}{dt} = \frac{\partial V}{\partial r} \frac{dr}{dt} + \frac{\partial V}{\partial h} \frac{dh}{dt}$ to find $\frac{dV}{dt}$ when $r = 5$, $h = 10$.
- 20 The area of a triangle with sides a , b and included angle θ is $A = \frac{1}{2}ab \sin \theta$. If $a = 5$ cm is fixed, b is increasing at 2 cm/s, and $\theta = \frac{\pi}{6}$ is increasing at 0.1 rad/s, find $\frac{dA}{dt}$ when $b = 8$.
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The gradient in three dimensions

- 21 Find ∇f for $f(x, y, z) = x^2y + yz^2$ at the point $(1, 2, 3)$.
- 22 Find the directional derivative of $f(x, y, z) = xyz$ at $(1, 1, 1)$ in the direction of $\mathbf{v} = \langle 1, 1, 1 \rangle$.
- 23 Find the direction of steepest ascent of $f(x, y, z) = x^2 + y^2 + z^2$ at $(1, 2, 2)$, and the maximum rate of increase.
- 24 Explain briefly why ∇f for $f(x, y, z) = x^2 + y^2 + z^2$ always points radially outward from the origin.

- 25 For $f(x, y, z) = xy + yz + zx$, find ∇f at $(1, -1, 2)$ and the directional derivative in the direction of $\mathbf{v} = \langle 0, 0, 1 \rangle$.